Predicting Coronary Heart Disease from Lifestyle Survey Data with Regularized Logistic Regression

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Abstract—We predict whether a respondent of the 2015 Behavioral Risk Factor Surveillance System (BRFSS) has been diagnosed with coronary heart disease or myocardial infarction (MICHD) from 321 survey answers. Using only NumPy, we compare linear and logistic regression baselines and build a feature pipeline that one-hot encodes categorical answers, keeps missingness as information, cleans survey answer codes and discretizes numeric variables. Combined with L_2-regularized logistic regression solved by Newton's method and an F1-optimal decision threshold, our model reaches an F1 score of 0.440 ± 0.008 in 5-fold cross-validation. An ablation study shows that choosing the decision threshold matters far more than any feature engineering step: it raises F1 from 0.27 to 0.44.

I. Introduction

Cardiovascular diseases are a leading cause of death worldwide, and identifying at-risk individuals from cheap survey data could support prevention. The task is binary classification: given a respondent's answers, predict MICHD ($y = 1$) or not ($y = -1$). The data contain 328,135 training and 109,379 test respondents. Only 8.8% of training respondents are positive, and the competition metric is the F1 score of the positive class. Both facts shape our design choices.

II. Data and Feature Engineering

Exploratory analysis. Of the 321 columns, 239 contain missing values and 147 are more than half missing. This is mostly by design: BRFSS skips follow-up questions when they do not apply (e.g. diabetes questions for non-diabetics). 261 columns have at most 60 distinct values and encode categorical answers. Raw questions use special codes (e.g. 77 = "don't know", 99 = "refused", 88 = "none") that would distort a numeric treatment. The features most correlated with the label are general health, age, employment status (retired), high blood pressure and diabetes.

Pipeline. All statistics below are fitted on training data only and re-fitted inside every cross-validation fold.

1) Column filtering: we drop 18 survey bookkeeping columns (interview date, sequence numbers, sampling weights, phone screening) as well as constant columns.
2) Categorical columns (≤ 30 distinct values) are one-hot encoded, keeping every category that appears in $\geq 0.1\%$ of training rows. We add an indicator for "missing" because a skipped question is informative.
3) Numeric columns (> 30 values): in raw questions, "none" codes become 0 and "don't know/refused" codes become missing. A code is cleaned only if it lies above every regular answer of that question, so that e.g. 7 days of exercise is not removed. Missing values are imputed with the training median and flagged with an indicator. Values are clipped to the $[0.5, 99.5]$ percentiles and standardized. We also add indicators for 10 quantile bins, which lets the linear model learn a piecewise-constant, non-monotone effect of e.g. age or BMI.

The final design matrix has 1,733 features plus a bias column.

III. Models and Training

Basic methods. We implemented the six required methods: least squares via the normal equations, ridge regression, MSE gradient descent and SGD, and (regularized) logistic regression with gradient descent. We use ridge regression (on targets in $\{-1, 1\}$) and gradient-descent logistic regression as baselines.

Final model. We minimize the regularized logistic loss

$$\mathcal{L}(w) = \frac{1}{N} \sum_{i=1}^{N} \left[\log\left(1 + e^{x_i^\top w}\right) - y_i x_i^\top w \right] + \lambda \|w\|^2$$

with $y_i \in \{0, 1\}$, leaving the bias unpenalized. Plain gradient descent converges slowly on 1,700 correlated indicator features. We therefore use Newton's method, $w \leftarrow w - t\, H^{-1} \nabla \mathcal{L}$ with $H = \frac{1}{N} X^\top \mathrm{diag}(p_i(1-p_i)) X + 2\lambda I$, and a backtracking line search on t. It converges in about 8 iterations ($\sim$10 s on the full data). Regularization is necessary: with $\lambda = 0$ the Hessian is singular because the one-hot blocks are collinear.

Decision threshold. With 8.8% positives, the usual cut-off $p > 0.5$ predicts very few positives and gives poor F1. We sort the training scores and pick the threshold that maximizes training F1 exactly; it is $p \approx 0.21$.

Evaluation protocol. We use 5-fold cross-validation with the same random folds for every configuration. In each fold, the feature pipeline, the model and the

threshold are all fitted on the four training folds and evaluated once on the held-out fold. We report the mean (and standard deviation) of F1 over folds, together with accuracy and ROC AUC.

IV. Results

Table I starts from baselines and adds one modification at a time. Raw features are median-imputed, code-cleaned and standardized original columns.

Table I
5-fold CV results. All rows except the last use the tuned threshold.

Configuration	F1	Acc.	AUC
Ridge regression, raw	0.429	0.875	0.858
Logistic, GD (300 it., γ=0.5), raw	0.416	0.864	0.850
Logistic, Newton, raw	0.436	0.877	0.864
+ drop admin columns	0.436	0.880	0.864
+ one-hot categoricals	0.439	0.876	0.866
+ quantile bins (final)	0.440	0.880	0.865
Ridge regression, final features	0.432	0.876	0.858
Final model, threshold $p = 0.5$	0.267	–	–

Threshold. Moving the threshold from 0.5 to the F1-optimal value is by far the most important choice (+0.17 F1). On a held-out 20% split, recall rises from 0.17 to 0.55 while precision falls from 0.60 to 0.37. Accuracy drops from 0.916 to 0.875, which shows why accuracy alone is misleading under class imbalance.

Optimizer. With the same objective, 300 steps of gradient descent reach only F1 0.416, compared with 0.436 for Newton's method. The gap comes from incomplete optimization, not from the model.

Features. One-hot encoding and binning add +0.004 F1 and +0.001 AUC over the raw logistic model. This is consistent across folds but comparable to the fold-to-fold standard deviation ($\approx$ 0.006). The raw features already capture most of the signal because many BRFSS variables are ordinal with a roughly monotone risk. Dropping bookkeeping columns barely changes performance, but it makes the model rely on health information rather than on survey logistics.

Logistic vs. least squares. With identical features, logistic regression beats ridge regression by 0.008 F1 and 0.007 AUC. Squared loss penalizes confidently correct predictions and is therefore a poor surrogate for classification under class imbalance.

Regularization. Sweeping $\lambda \in \{10^{-6}, \ldots, 10^{-2}\}$, F1 is flat (0.439–0.440) for $\lambda \in [10^{-5}, 10^{-4}]$. It decreases for stronger penalties (0.436 at 10^{-3}, 0.422 at 10^{-2}). We use $\lambda = 10^{-4}$, which has the best AUC. With 1,700 features and over 260k training rows, the model is far from overfitting: training F1 (0.449) is close to validation F1.

Extensions that did not help. We tried several extensions of the final model, none of which improved F1 beyond noise. With 5-fold CV (per-fold differences to the final model in parentheses), we tested choosing the threshold on inner out-of-fold scores instead of training scores (+0.0008; folds from -0.002 to +0.003), treating columns with up to 70 values as categorical (+0.0005), averaging models with 5, 10 and 20 bins (+0.0003), and bagging 5 bootstrap models (+0.0002). On a held-out 20% split, we also added 120 interaction features (age and sex times the 60 largest-weight features; F1 unchanged at 0.440, AUC +0.0007) and trained a one-hidden-layer ReLU network with Adam (64 units: 0.437; 128–256 units with dropout: 0.41–0.42, overfitting). Averaging the network with logistic regression gave 0.441, again within noise.

Test set. Trained on all data, the model predicts 12.7% of test respondents as positive. On the competition leaderboard it scores F1 = 0.447 and accuracy = 0.882, consistent with our cross-validation estimate (0.440 ± 0.008).

V. Discussion

A well-optimized linear model with careful handling of survey codes and missingness gives a strong baseline. Most of the achievable F1 comes from aligning the decision rule with the metric rather than from model complexity. Ranking quality stays at AUC $\approx$ 0.865 across every feature set, the interaction features and a neural network, which suggests the remaining error is largely irreducible given these self-reported answers. Gradient-boosted trees are the main model family we did not try.